Statistical Distributions and Testing

Questions 1–123

1.55 — Question 1. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	0.11	0.13	0.29	0.30	0.17

Find $\mathrm{P}(2 \leq X < 5)$

1.55 — Question 2. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x}{100} + \frac{1}{5}$$

for $x = 2, 4, 6, 8$. Draw a table showing the probability distribution of X.

1.55 — Question 3. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x + 6}{40}$$

for $x = 1, 3, 5, 7$. Find $\mathrm{P}(X > 4)$

1.55 — Question 4. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3	4 or more
$\mathrm{P}(X = x)$	0.05	0.1	k	0.35	0.32

Find the value of k

1.55 — Question 5. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4	5
$\mathrm{P}(X = x)$	k	$3k$	$5k$	k	k

Find the value of k

1.55 — Question 6. A probability distribution is given by

$$\mathrm{P}(X = x) = k(10 - x)$$

for $x = 2, 4, 6, 8$ where k is a constant.

a) Show that $k = \frac{1}{20}$

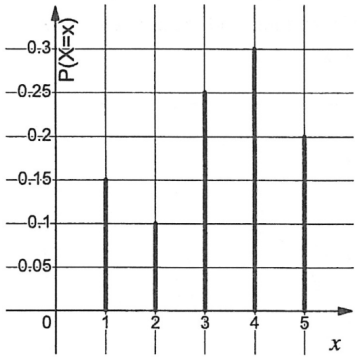
b) Calculate $\mathrm{P}(X \geq 6)$

1.55 — Question 7. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{x + k}{10x + 10}$$

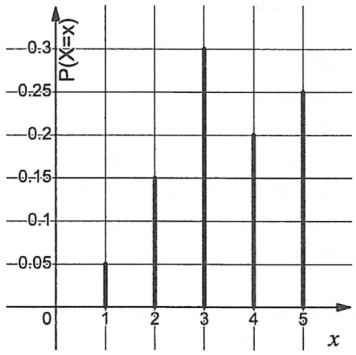
for $x = 1, 2, 3$. Find the value of k

1.55 — Question 8. The stick graph below shows the probability distribution for the discrete random variable X.



Find $\mathrm{P}(1 < X \leq 4)$

1.55 — Question 9. The stick graph below shows the probability distribution for the discrete random variable X.



Two values of X are selected at random. Assume these events are independent. Find the probability that both values are greater than or equal to 4

1.55 — Question 10. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.71	0.19	0.1

Two values of X are chosen at random. Assume these events are independent. Find the probability that both values are the same.

1.55 — Question 11. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.9	0.09	0.01

Two values of X are chosen at random. Assume these events are independent. Find the probability that the second value is less than the first.

1.55 — Question 12. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3
$\mathrm{P}(X = x)$	0.24	0.36	0.4

Two values of X are chosen at random. Assume these events are independent. Find the probability that the first value is odd and the second value is prime.

1.55 — Question 13. The table below shows the probability distribution for a discrete random variable X.

x	5	8	11
$\mathrm{P}(X = x)$	0.15	0.25	0.6

Two values of X are chosen at random. Assume these events are independent. Find the probability that the sum of the two scores is less than 17

1.55 — Question 14. In a game, one spin of a biased spinner can score 1, 2, 3 or 4 points. The random variable X represents a player's score, and this has the following probability distribution:

x	1	2	3	4
$\mathrm{P}(X = x)$	0.02	0.14	0.26	0.58

The spinner is spun twice and the scores added. It is given that each game is independent. Find the probability of getting a total of 4 points.

1.55 — Question 15. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2
$\mathrm{P}(X = x)$	0.33	0.22	0.45

Two values of X are chosen at random. Find the probability that the product of these two values is zero.

1.55 — Question 16. The discrete random variable X is modelled by the probability distribution:

$$\mathrm{P}(X = x) = \frac{x}{20}$$

for $x = 3, 4, 6$ and 7. The independent random variables X_1 and X_2 have the same distribution as X. Find $\mathrm{P}(X_1 = X_2)$

1.55 — Question 17. A spinner can be modelled by X with the probability distribution:

$$\mathrm{P}(X = x) = \frac{x+1}{14}$$

for $x = 1, 2, 3$ and 4. The spinner is spun three times. Find the probability that the first value is greater than the sum of the next two values.

1.55 — Question 18. It is given that a discrete random variable X can take on the values 1, 2, 4 and 5, and that $\mathrm{P}(X = 1) = 0.2$. When $r = 1$ and $r = 2$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 3)$
Draw a table to show the probability distribution of X.

1.55 — Question 19. It is given that a discrete random variable X can take on the values 4, 5, 9 and 10, and that $\mathrm{P}(X = 5) = 0.18$. When $r = 4$ and $r = 5$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 5)$
Draw a table to show the probability distribution of X.

1.55 — Question 20. It is given that a discrete random variable X can take on the values 2, 3, 4, 5 and 6, and that $\mathrm{P}(X = 2) = 0.11$. When $r = 2$ and $r = 3$, it is also given that $\mathrm{P}(X = r) = \mathrm{P}(X = r + 3)$
It is also given that $\mathrm{P}(X = 4) = \mathrm{P}(X = 3) + \mathrm{P}(X = 5)$. Draw a table to show the probability distribution of X.

1.55 — Question 21. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} \frac{1}{5} \times \frac{2^{x-1}}{3^{x-1}} & x = 1, 2 \\ k & x = 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

a) Find $\mathrm{P}(X = 2)$.
b) Find the value of k
c) Find $\mathrm{P}(2 \leq X \leq 3)$

1.55 — Question 22. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} ax^2 & x = 1, 2 \\ bx + 0.05 & x = 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants.

a) Find $\mathrm{P}(X = 2)$ in terms of a
b) Given that $\mathrm{P}(X = 2) = \mathrm{P}(X \geq 3)$, find the exact values of a and b

1.55 — Question 23. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X = x) = \begin{cases} a(15 - 3x) & x = 1, 2, 3, 4 \\ bx & x = 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where a and b are constants.

a) Show that $30a + 11b = 1$

b) Given that $\mathrm{P}(X \geq 4) = \frac{1}{4}$ find the exact values of a and b

1.55 — Question 24. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3
$\mathrm{P}(X = x)$	$3k^2$	$3k$	$\frac{9}{2}k^2$	$\frac{1}{2}k$

Find the value of k and hence find $\mathrm{P}(X > 1)$

1.55 — Question 25. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X=x)=\begin{cases}k & x=1,2,3,4\\ \frac{1}{3}\mathrm{P}(X=x-1) & x=5,6\\ 0 & \text{otherwise}\end{cases}$$

where k is a constant.
Show that $k=\frac{9}{40}$

1.55 — Question 26. The discrete random variable X is modelled by the following probability distribution:

$$\mathrm{P}(X=x)=\begin{cases}\frac{1}{5}\mathrm{P}(X=x+1) & x=1,2,3\\ k & x=4,5\\ 0 & \text{otherwise}\end{cases}$$

where k is a constant. Find the value of k and hence find $\mathrm{P}(X\leq 3)$

1.55 — Question 27. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3
$\mathrm{P}(X = x)$	$2k$	0.42	0.37	k

Two values of X are chosen at random. Find the probability that the product of these two values is greater than their sum.

1.55 — Question 28. The table below shows the probability distribution for a discrete random variable X.

x	0	1	2	3	4
$\mathrm{P}(X = x)$	$\frac{1}{9}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{5}{18}$

Two independent values, X_1 and X_2, of X are chosen. Find $\mathrm{P}(X_1 + X_2 = 5)$

1.55 — Question 29. The table below shows the probability distribution for a discrete random variable X.

x	2	3	4	5
$\mathrm{P}(X = x)$	0.8	0.09	0.06	0.05

It is given that $Y = X^2$. Find $\mathrm{P}(Y + 2X \leq 15)$

1.55 — Question 30. The table below shows the probability distribution for a discrete random variable X.

x	4	5	9	13
$\mathrm{P}(X = x)$	0.7	0.2	0.05	0.05

It is given that $Y = \frac{15}{X}$. Find $\mathrm{P}(X + Y \geq 8)$

1.55 — Question 31. Tom models the number of games of pool he will win in a match by the random variable X with the following probability distribution:

x	0	1	2	3	More than 3
$\mathrm{P}(X = x)$	$\frac{1}{15}$	$\frac{1}{10}$	$\frac{1}{3}$	$\frac{1}{2}$	0

The number of games of pool won in a match is independent of any other match. Tom chooses three pool matches at random. Use the model to find the probability of:

a) Tom winning the same number of games of pool in all three matches;
b) Tom winning exactly two games of pool in total in the three matches;
c) Tom winning fewer games of pool than the preceding match across the three matches.

1.55 — Question 32. The table below shows the probability distribution for a discrete random variable X.

x	1	2	3	4
$\mathrm{P}(X = x)$	a	b	c	0.45

It is given that $\mathrm{P}(X \leq 2) = \frac{1}{3}\mathrm{P}(X \geq 3)$.
Find $\mathrm{P}(X \geq 3)$

1.55 — Question 33. A probability distribution is given by

$$\mathrm{P}(X = x) = \begin{cases} \frac{k}{x+1} & x = 1, 3, 7 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant. Three independent values, X_1, X_2 and X_3, of X are chosen.
Find $\mathrm{P}(X_1 + X_2 > X_3)$

1.55 — Question 34. The table below shows the probability distribution for a discrete random variable S.

s	6	12	18	24
$\mathrm{P}(S = s)$	$3k$	$4k$	$5k$	$6k$

S_1 and S_2 have the same distribution as S and are independent. Find $\mathrm{P}(S_1 + S_2 = 36)$

1.55 — Question 35. Two bias spinners, X and Y, can be represented by the probability distributions given by $\mathrm{P}(X = x) = \frac{x}{25}$ for $x = 12, 13$, and $\mathrm{P}(Y = y) = \frac{y}{27}$ for $x = 13, 14$ respectively. Spinner X is spun first and then spinner Y is spun. Find the probability that the sum of the scores is 26

1.55 — Question 36. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values 1, 2, 3, 4 and 5. Two values, X_1 and X_2, of X are chosen at random. Find $\mathrm{P}(X_1 = X_2)$

1.55 — Question 37. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values $1, 2, 3, \ldots, n$. Two values, X_1 and X_2, of X are chosen at random. Find $\mathrm{P}(X_1 = X_2)$

1.55 — Question 38. The random variable X is a discrete uniform distribution where every outcome is equally likely. X can take on the values $1, 2, 3, \ldots, n$. Two values, X_1 and X_2, of X are chosen at random. It is given that $\mathrm{P}(X_1 = X_2) = 0.1$. Find $\mathrm{P}(X_1 < X_2)$

1.55 — Question 39. Prerequisite knowledge for Q39-Q40: Conditional Probability A probability distribution is given by

$$\mathrm{P}(X = x) = \begin{cases} 2kx & x = 3, 4, 5, 6 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant. Two independent values, X_1 and X_2, of X are chosen.

a) Find $\mathrm{P}(X_1 + X_2 = 9)$

b) It is given that $X_1 + X_2 = 9$. Find the probability that one of X_1 or X_2 is 4

1.55 — Question 40. A probability distribution is given by

$$\mathrm{P}(X = x) = \frac{2^x}{k}$$

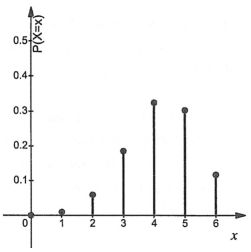
for $x = 2, 3, 4, 5$ where k is a constant. Three independent values, X_1, X_2 and X_3, of X are chosen.

a) Find $\mathrm{P}(X_1 + X_2 + X_3 = 9)$

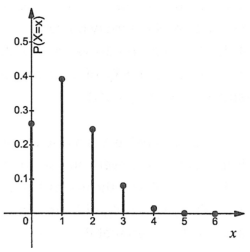
b) It is given that $X_1 + X_2 + X_3 = 9$. Find the probability that at least one of X_1, X_2 or X_3 is 4

1.56 — Question 1. The four diagrams below show $X \sim \mathrm{B}(6, 0.2)$, $X \sim \mathrm{B}(6, 0.5)$, $X \sim \mathrm{B}(6, 0.7)$ and $X \sim \mathrm{B}(6, 0.89)$.
Which is which?

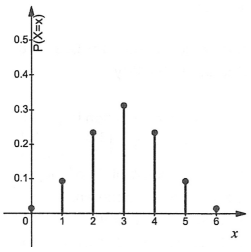
A



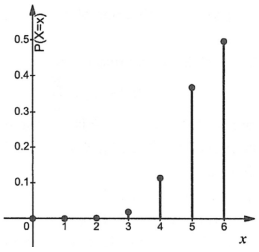
B



C



D



1.56 — Question 2. Consider the stick graph of the following distributions. Do they show a positive skew, a negative skew, or are they symmetric?

a) $X \sim \mathrm{B}(45, 0.3)$
b) $X \sim \mathrm{B}(32, 0.5)$
c) $X \sim \mathrm{B}(24, 0.9)$
d) $X \sim \mathrm{B}(60, 0.51)$

1.56 — Question 3. Given $X \sim \mathrm{B}(3, 0.2)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3
$\mathrm{P}(X = x)$				

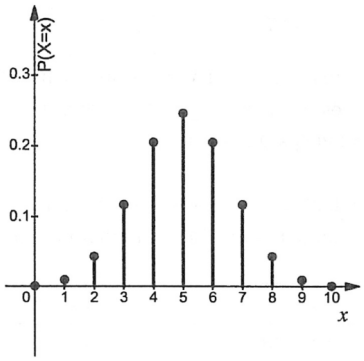
1.56 — Question 4. Given $X \sim \mathrm{B}(3, 0.6)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3
$\mathrm{P}(X = x)$				

1.56 — Question 5. Given $X \sim \mathrm{B}(4, 0.5)$, copy and complete the table shown below and draw a stick graph to represent the distribution.

x	0	1	2	3	4
$\mathrm{P}(X = x)$					

1.56 — Question 6. The diagram below shows $X \sim \mathrm{B}(n, p)$. Write down n and suggest a value for p



1.56 — Question 7. Given $X \sim \mathrm{B}(15, 0.3)$, **use the formula** to find:

a) $\mathrm{P}(X = 3)$
b) $\mathrm{P}(X = 4)$
c) $\mathrm{P}(X = 5)$
d) $\mathrm{P}(X = 9)$

1.56 — Question 8. Use the formula in each case.

a) Given $X \sim \mathrm{B}(10, 0.6)$, find $\mathrm{P}(X = 6)$
b) Given $X \sim \mathrm{B}(25, 0.2)$, find $\mathrm{P}(X = 4)$
c) Given $X \sim \mathrm{B}(60, 0.9)$, find $\mathrm{P}(X = 57)$
d) Given $X \sim \mathrm{B}(105, 0.56)$, find $\mathrm{P}(X = 52)$

1.56 — Question 9. Given $X \sim \mathrm{B}(10, 0.7)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X = 5)$
b) $\mathrm{P}(X = 7)$
c) $\mathrm{P}(X = 10)$

1.56 — Question 10. Given $X \sim \mathrm{B}(20, 0.2)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X = 1)$
b) $\mathrm{P}(X = 3)$
c) $\mathrm{P}(X = 5)$

1.56 — Question 11. Given $X \sim \mathrm{B}(17, 0.16)$, **use the formula** to find:

a) $\mathrm{P}(X \leq 1)$
b) $\mathrm{P}(X \leq 2)$
c) $\mathrm{P}(X \leq 3)$

1.56 — Question 12. Given $X \sim \mathrm{B}(22, 0.35)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X \leq 2)$
b) $\mathrm{P}(X \leq 5)$
c) $\mathrm{P}(X \leq 8)$

1.56 — Question 13. Use your calculator's binomial probability function in each case.

a) Given $X \sim \mathrm{B}(18, 0.24)$, find $\mathrm{P}(X \leq 5)$
b) Given $X \sim \mathrm{B}(27, 0.36)$, find $\mathrm{P}(X \leq 15)$
c) Given $X \sim \mathrm{B}(39, 0.49)$, find $\mathrm{P}(X \leq 20)$
d) Given $X \sim \mathrm{B}(47, 0.61)$, find $\mathrm{P}(X \leq 25)$

1.56 — Question 14. Given $X \sim \mathrm{B}(18, 0.68)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X < 9)$
b) $\mathrm{P}(X < 11)$
c) $\mathrm{P}(X < 14)$

1.56 — Question 15. Given $X \sim \mathrm{B}(17, 0.49)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(X \geq 8)$

b) $\mathrm{P}(X \geq 11)$

c) $\mathrm{P}(X \geq 12)$

1.56 — Question 16. Use your calculator's binomial probability function in each case.

a) Given $X \sim \mathrm{B}(16, 0.37)$, find $\mathrm{P}(X \geq 4)$

b) Given $X \sim \mathrm{B}(38, 0.02)$, find $\mathrm{P}(X \geq 2)$

c) Given $X \sim \mathrm{B}(71, 0.77)$, find $\mathrm{P}(X \geq 52)$

1.56 — Question 17. Given $X \sim \mathrm{B}(23, 0.66)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(9 \leq X \leq 12)$
b) $\mathrm{P}(10 \leq X \leq 14)$
c) $\mathrm{P}(12 \leq X \leq 14)$

1.56 — Question 18. Given $X \sim \mathrm{B}(18, 0.24)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(2 < X \leq 3)$
b) $\mathrm{P}(3 < X \leq 7)$
c) $\mathrm{P}(4 < X \leq 10)$

1.56 — Question 19. Given $X \sim \mathrm{B}(29, 0.49)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(14 \leq X < 17)$

b) $\mathrm{P}(13 \leq X < 18)$

c) $\mathrm{P}(10 \leq X < 13)$

1.56 — Question 20. Given $X \sim \mathrm{B}(13, 0.45)$, use your calculator's binomial probability function to find:

a) $\mathrm{P}(4 < X < 6)$

b) $\mathrm{P}(5 < X < 10)$

c) $\mathrm{P}(6 < X < 9)$

1.56 — Question 21. In a large population of a certain type of tropical bird, 20% are known to have yellow feathers. A random sample of 30 of these birds are selected. Calculate the probability that the sample contains exactly 5 with yellow feathers.

1.56 — Question 22. A factory produces blades for pencil sharpeners. The probability of one of these blades being misshapen is 0.008. In a random sample of 200 blades, find the probability of less than 3 being misshapen.

1.56 — Question 23. Write down the mean, $\mathrm{E}(X)$, and variance, $\mathrm{Var}(X)$ of the following distributions.

a) $X \sim \mathrm{B}(8, 0.5)$
b) $X \sim \mathrm{B}(20, 0.3)$
c) $X \sim \mathrm{B}(16, 0.9)$
d) $X \sim \mathrm{B}(25, 0.2)$
e) $X \sim \mathrm{B}(40, 0.6)$

1.56 — Question 24. Given $X \sim \mathrm{B}(22, 0.6)$ and the mean, $\mathrm{E}(X) = \mu$,

a) find $\mathrm{P}(X \leq \mu)$,
b) find $\mathrm{P}(X > \mu)$.

1.56 — Question 25. Given $X \sim \mathrm{B}(n, p)$ and $\mathrm{P}(X = 10) = \binom{45}{10} \times 0.25^{10} \times 0.75^{35}$ write down the mean, $\mathrm{E}(X)$, and variance, $\mathrm{Var}(X)$.

1.56 — Question 26. Given $X \sim \mathrm{B}(450, p)$ and the mean, $\mathrm{E}(X)$, is 100, find the value of p

1.56 — Question 27. Given $X \sim \mathrm{B}(1560, p)$ and the mean, $\mathrm{E}(X)$, is 120, find the value of p

1.56 — Question 28. Given $X \sim \mathrm{B}(n, 0.3)$ and the variance, $\mathrm{Var}(X)$, is 42, find the value of n

1.56 — Question 29. Given $X \sim \mathrm{B}(n, \frac{1}{6})$ and the variance, $\mathrm{Var}(X)$, is 375, find the value of n

1.56 — Question 30. Given $X \sim \mathrm{B}(n, \frac{1}{8})$ and the standard deviation is 35, find the value of n

1.56 — Question 31. Given $X \sim \mathrm{B}(n, \frac{5}{8})$ and the standard deviation is 105, find the value of n

1.56 — Question 32. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 25.2$ and $\mathrm{Var}(X) = 2.52$, find the values of n and p

1.56 — Question 33. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 14.4$ and $\mathrm{Var}(X) = 8.64$, find the values of n and p

1.56 — Question 34. Given $X \sim \mathrm{B}(n, p)$, with $\mathrm{E}(X) = 5$ and $\mathrm{Var}(X) = \frac{60}{13}$, find the values of n and p

1.56 — Question 35. Given $X \sim \mathrm{B}(100, 0.2)$ and $Y \sim \mathrm{B}(20, k)$, and that $\mathrm{Var}(X) = \mathrm{E}(Y)$, find the value of k

1.56 — Question 36. It is given that $X \sim \mathrm{B}(n, p)$ and $Y \sim \mathrm{B}(k, p)$, where $0 < p < 1$. It is also given that $\mathrm{E}(X) = 3\mathrm{E}(Y)$ and $\mathrm{Var}(X) = 2\mathrm{E}(Y)$. Find the value of p

1.56 — Question 37. It is given that $X \sim \mathrm{B}(n, p)$ and $Y \sim \mathrm{B}(k, p)$, where $0 < p < 1$. It is also given that $\mathrm{E}(X) = 5\,\mathrm{Var}(Y)$ and $\mathrm{E}(Y) = 3\mathrm{E}(X)$. Find the value of p

1.56 — Question 38. State the conditions under which a situation can be modelled by a binomial distribution.

1.56 — Question 39. A fair six-sided die is rolled until a 6 appears. Explain why we cannot use a binomial distribution to model this.

1.56 — Question 40. In a small village, the probability of it raining on any one day of the year is known to be 0.22. Nico has sixteen days between breaking up and going back to school for his Easter holidays and wants to determine the probability that for at most 3 days it will be raining. Explain why it would be inappropriate to use a binomial distribution in this situation.

1.56 — Question 41. Georgia attempts a crossword puzzle on the puzzle page of a newspaper every day. Over several months, Georgia has found that she is able to successfully complete the crossword puzzle 82% of the times.

a) Using the binomial distribution, find the probability that, out of a random sample of 21 days, Georgia successfully completes at least 18 crossword puzzles.

b) Using the same binomial distribution in part a), find the probability that Georgia successfully completes less than 16 crossword puzzles.

1.56 — Question 42. A fair 8-sided spinner has sides numbered 1, 2, 3, 4, 5, 6, 7 and 8. The spinner is spun 35 times. The random variable X represents the number of times the spinner lands on 8.

a) Find the variance, $\text{Var}(X)$, of the number of times the spinner lands on 8

b) Find the probability that the spinner lands on an 8 at most 6 times.

c) Find $\text{P}(4 < X < 8)$.

1.56 — Question 43. Emma plays snooker and is practicing a trick-shot. Each evening, she attempts the trick-shot 50 times. Every time she attempts the trick-shot, there is a probability of 0.1 that she miss-cues. The situation is modelled as a binomial distribution.

a) Find the mean number of times that Emma miss-cues in one evening.
b) Find the variance of the number of times that Emma miss-cues in one evening.
c) Find the probability that, on a particular evening, she miss-cues exactly 5 times.
d) Find the probability that, on a particular evening, she miss-cues at most 6 times.
e) Emma practices the trick-shot 50 times each evening for one week. Find the probability that she miss-cues at most 6 times on each of the 7 days.

1.56 — Question 44. Curtis plays darts. He has determined the probability of him hitting the bullseye with one dart is 0.15. Curtis throws twelve darts. The situation is modelled as a binomial distribution.

a) Find the probability that Curtis will hit the bullseye more than twice.
b) Curtis throws twelve darts at three different randomly selected times. Find the probability that Curtis will hit the bullseye more than twice on one of these three times.
c) State two assumptions, in context, that would be necessary for the binomial distribution to be valid in this case.

1.56 — Question 45. In a survey of 3500 A-Level Maths students across several sixth form colleges, 2240 use graphical calculators.

a) Using the binomial distribution, find the probability that, out of a random sample of 24 students, exactly 15 students have a graphical calculator.

b) Explain why the answer to part a) may not be valid for 24 students from a single class.

1.56 — Question 46. A shop sells interlocking floor tiles. Tara visits the shop to purchase thirty of these tiles. It is given that 2.5% of all interlocking floor tiles of these style are faulty. By modelling the situation as a binomial distribution,

a) Find the probability that none of the tiles Tara buys are faulty.

b) Find the probability that at least three of the tiles Tara buys are faulty.

c) State two assumptions, in context, that would be necessary for the binomial distribution to be valid in this case.

1.56 — Question 47. At a small sixth form college, 12% of the students belong to the cinema club. A random sample of 40 students are selected from the college. By modelling the situation as a binomial distribution,

a) Find the probability that exactly 4 students belong to the cinema club.
b) Find the probability that at least 3 but at most 8 belong to the cinema club.
c) Of the students who belong to the cinema club, 30% also belong to the book club. Assuming that these events are independent, find the probability that a randomly chosen student belongs to both the cinema club and the book club.
d) In the random sample of 40 students, find the probability that exactly one student belongs to both the cinema club and the book club.

1.56 — Question 48. A type of lightbulb has a $\frac{3}{16}$ chance of being faulty. In a random sample of 15 lightbulbs, show that the probability of 2 lightbulbs being faulty is the same as the probability of 3 lightbulbs being faulty.

1.56 — Question 49. Every day, Kai takes a walk in a park near his house. The probability that he spots a squirrel in the park on any particular day is 0.13

a) Find the probability that in a week Kai spots a squirrel on exactly 3 days.
b) Find the probability that Kai spots a squirrel on exactly 3 days in a week, during at most 1 of 7 randomly chosen weeks.

1.56 — Question 50. Prerequisite knowledge for Q50: Conditional Probability A fair die has eight faces numbered 1 to 8
The die is rolled 40 times and the number of times an 8 is rolled is recorded. Find the probability that at least eight 8s are rolled,

a) given that fewer than eleven 8s are rolled.
b) given that more than five 8s are rolled.
c) given that more than five but fewer than eleven 8s are rolled.

1.56 — Question 51. On each of the twelve floors of a large office building, there is a printer. Four of these are designated to only print colour, and the other eight only print black and white. The colour printers are in use 15% of the time and the black and white printers are in use 35% of the time. Find the probability that at any one time,

a) one of the colour printers is in use;
b) at most two black and white printers are in use;
c) none of the printers are in use;
d) exactly two of the printers are in use.

1.56 — Question 52. Given $X \sim \mathrm{B}(150, 0.25)$, find the largest possible value of k such that $\mathrm{P}(X \leq k) < 0.1$

1.56 — Question 53. Given $X \sim \mathrm{B}(500, 0.15)$, find the largest possible value of k such that $\mathrm{P}(X < k) < 0.1$

1.56 — Question 54. In a town, it is known that 10% of local households use online food shopping. A random sample of 300 households are selected. The random variable X is defined as the number of households in the sample who use online food shopping. Calculate the largest possible value of k such that $\mathrm{P}(X < k) < 0.01$

1.56 — Question 55. In a constituency, it is known that 35% of constituents traditionally vote for the independent candidate. A random sample of 400 adults in the constituency are selected. The random variable X is defined as the number of adults in the sample who voted for the independent candidate. Calculate the least possible value of k such that $\mathrm{P}(X > k) < 0.05$

1.56 — Question 56. Prerequisite knowledge for Q56–Q59: Logarithms It is given that $X \sim \mathrm{B}(n, 0.07)$. Estimate the value of n given that $\mathrm{P}(X = 0) = 0.00465$

1.56 — Question 57. It is given that $X \sim \mathrm{B}(n, 0.03)$. Estimate the value of n given that $\mathrm{P}(X = 0) = 0.02921$

1.56 — Question 58. It is given that $X \sim \mathrm{B}(n, 0.995)$. Estimate the value of n given that $\mathrm{P}(X = n) = 0.03983$

1.56 — Question 59. It is given that $X \sim \mathrm{B}(n, 0.974)$. Estimate the value of n given that $\mathrm{P}(X = n) = 0.01557$
Use your calculator's SOLVE function for Q60 and Q61

1.56 — Question 60. It is given that $X \sim \mathrm{B}(n, 0.023)$. Estimate the value of n given that $\mathrm{P}(X \leq 1) = 0.39021$

1.56 — Question 61. It is given that $X \sim \mathrm{B}(n, 0.998)$. Estimate the value of n given that $\mathrm{P}(X \geq n-1) = 0.72913$

1.56 — Question 62. Prerequisite knowledge for Q62–Q63: Logarithms In a town, it is known that one person in every 15 people, on average, has the blood type B positive. A local hospital is looking for potential donors of this blood type. Determine the least number of blood donors that must be selected, at random, in order for the probability of including at least one donor that is type B positive is 0.95 or more.

1.56 — Question 63. A factory has been made aware that there is a fault that affects one in every 70 components that it manufactures. To investigate, an employee is tasked with taking a random sample of components from the production line and finding a faulty component.
Determine the least number of components that must be randomly sampled in order for the probability of including at least one faulty component is 0.99 or more.

1.57 — Question 1. Describe what is meant by the significance level.

1.57 — Question 2. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.7$, $H_1 : p < 0.7$
$X \sim B(20, 0.7)$
Observed 10, 5% significance level

b) $H_0 : p = 0.3$, $H_1 : p < 0.3$
$X \sim B(30, 0.3)$
Observed 5, 5% significance level

c) $H_0 : p = 0.6$, $H_1 : p < 0.6$
$X \sim B(27, 0.6)$
Observed 12, 10% significance level

d) $H_0 : p = 0.9$, $H_1 : p < 0.9$
$X \sim B(52, 0.9)$
Observed 40, 1% significance level

1.57 — Question 3. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.4$, $H_1 : p > 0.4$
$X \sim B(15, 0.4)$
Observed 9, 5% significance level

b) $H_0 : p = 0.85$, $H_1 : p > 0.85$
$X \sim B(45, 0.85)$
Observed 41, 5% significance level

c) $H_0 : p = 0.55$, $H_1 : p > 0.55$
$X \sim B(36, 0.55)$
Observed 26, 10% significance level

d) $H_0 : p = 0.12$, $H_1 : p > 0.12$
$X \sim B(90, 0.12)$
Observed 18, 1% significance level

1.57 — Question 4. In each case, find the p-value and determine whether you reject H_0 or fail to reject H_0

a) $H_0 : p = 0.8$, $H_1 : p \neq 0.8$
$X \sim B(25, 0.8)$
Observed 15, 5% significance level

b) $H_0 : p = 0.72$, $H_1 : p \neq 0.72$
$X \sim B(33, 0.72)$
Observed 29, 5% significance level

c) $H_0 : p = 0.38$, $H_1 : p \neq 0.38$
$X \sim B(67, 0.38)$
Observed 33, 10% significance level

d) $H_0 : p = 0.05$, $H_1 : p \neq 0.05$
$X \sim B(180, 0.05)$
Observed 3, 1% significance level

1.57 — Question 5. Describe what is meant by:

a) the critical region,

b) the acceptance region.

1.57 — Question 6. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.5$, $H_1 : p < 0.5$
$X \sim B(18, 0.5)$
5% significance level

b) $H_0 : p = 0.55$, $H_1 : p < 0.55$
$X \sim B(38, 0.55)$
5% significance level

c) $H_0 : p = 0.35$, $H_1 : p < 0.35$
$X \sim B(29, 0.35)$
10% significance level

d) $H_0 : p = 0.86$, $H_1 : p < 0.86$
$X \sim B(65, 0.86)$
1% significance level

1.57 — Question 7. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.2$, $H_1 : p > 0.2$
$X \sim B(40, 0.2)$
5% significance level

b) $H_0 : p = 0.33$, $H_1 : p > 0.33$
$X \sim B(49, 0.33)$
5% significance level

c) $H_0 : p = 0.11$, $H_1 : p > 0.11$
$X \sim B(65, 0.11)$
10% significance level

d) $H_0 : p = 0.74$, $H_1 : p > 0.74$
$X \sim B(125, 0.74)$
1% significance level

1.57 — Question 8. In each case, find the critical region and write down the actual significance level.

a) $H_0 : p = 0.4$, $H_1 : p \neq 0.4$
$X \sim B(60, 0.4)$
5% significance level

b) $H_0 : p = 0.9$, $H_1 : p \neq 0.9$
$X \sim B(44, 0.9)$
5% significance level

c) $H_0 : p = 0.28$, $H_1 : p \neq 0.28$
$X \sim B(55, 0.28)$
10% significance level

d) $H_0 : p = 0.09$, $H_1 : p \neq 0.09$
$X \sim B(135, 0.09)$
1% significance level

1.57 — Question 9. Fran is a florist. She regularly makes the most sales the day before Mother's Day each year, and over several years 25% of customers are found to purchase a chrysanthemum bouquet on that day. Fran decides to make the chrysanthemum bouquets larger for the following year and consequently they become more expensive. She believes this will result in fewer bouquets of this variety being sold. The following year, in a random sample of 34 customers, 5 customers bought a chrysanthemum bouquet. Fran carries out a hypothesis test, at the 10% significance level, to test her belief. Fran's hypothesis test is shown below.

Let X be the number of customers purchasing a chrysanthemum bouquet.
$H_0 : p = 0.25$
$H_1 : p \leq 0.25$
$X \sim B(34, 0.25)$
$P(X = 5) = 0.0647$
$0.0647 > 0.1$ *so we fail to reject* H_0
There is insufficient evidence to suggest that the proportion of customers purchasing a chrysanthemum bouquet has decreased.

a) Identify the first two errors made by Fran in her hypothesis test.

b) Correct her test.

1.57 — Question 10. Chaisai is playing a game of chance that is designed to have a probability of winning of $\frac{1}{5}$. Chaisai believes he has found a way to improve his chances and wishes to conduct a hypothesis test to test this belief. He plays the game 45 more times and wins 15 times. Carry out a hypothesis test at the 5% significance level to test Chaisai's claim.

1.57 — Question 11. The National Health and Medical Research Council (NHMRC) 2013 Australian Dietary Guidelines set out the recommended servings of fruit and vegetables to be eaten per day. In 2022, it was reported that 44.1% of adults met the fruit recommendation of 2 servings. Amy believes that this proportion is too high. She takes a random sample of 60 adults and 20 of them reported they met the fruit recommendation of 2 servings a day. Carry out a hypothesis test at the 5% significance level to investigate whether Amy's belief is supported.

1.57 — Question 12. The UK employment rate for Sep-Nov 2023 was 75.8%. Daniel lives in Brighton and believes the employment rate there is significantly lower than the UK average. He takes a random sample of 110 adults between the ages of 16 to 64 and finds that 70 of them are employed. Carry out a hypothesis test at the 1% significance level to investigate whether Daniel's belief is supported.

1.57 — Question 13. Energee lithium-ion batteries have a 77% chance of lasting longer than 20 hours on a single charge. Energee releases a new version of the battery which they hope will on average last longer. A random sample of 150 of the new batteries is taken and 126 last longer than 20 hours. Carry out a hypothesis test at the 5% level of significance to investigate Energee's claim about the new batteries.

1.57 — Question 14. Thomas works for a company that designs and produces television screens. Thomas leads the team that develops the remote control, and it has been found that 4% of the remote controls produced have had a fault. A change is made to the manufacturing process and Thomas wants to determine if this has had any effect on changing the percentage that have a fault. A random sample of 60 brand new remote controls are checked and six are found to have the fault. Carry out a hypothesis test at the 5% significance level to investigate whether the new manufacturing process has changed the percentage that have a fault.

1.57 — Question 15. It is known that 35% of roses grown by a horticulturist suffer from powdery mildew. The horticulturist separates the roses from their other plants and grows them in a large new greenhouse instead. They want to conduct a hypothesis test to determine whether this has had any change on the proportion that suffer from powdery mildew. A random sample of 300 roses from the new greenhouse are collected and 81 are found to suffer from powdery mildew. Test, at the 3% significance level, whether there is any evidence to suggest the proportion has changed.

1.57 — Question 16. Chloe and Ajay are playing a board game with a single six-sided die. After several turns, Chloe believes the die is bias in favour of scoring a 1. They decide to pause the board game and roll the die 35 times to test this theory.

a) Write down suitable hypotheses to test Chloe's belief.
b) Using a 10% significance level, find the critical region for the test.
c) Of the 35 rolls, 1 appears seven times. Complete the test.

1.57 — Question 17. The probability of having to wait longer than three hours at a hospital emergency unit is known to be 76%. Another doctor is employed in the hope that waiting times will be reduced. A hypothesis test is to be conducted to test this belief.

a) Write down the hypotheses that should be used to test whether there has been a reduction in the number of patients waiting longer than three hours.
b) A random sample of 70 patients is taken. Using a 1% level of significance, find the critical region for the test.

Of the 70 patients in the sample, 43 waited longer than three hours to be seen.

c) Complete the test.

1.57 — Question 18. A large supermarket sells soup by the can or in packs of six. Of those customers who bought soup in 2022, it was recorded that 32% bought packs of six. During the following year, Becky, the manager of the supermarket, believes that the percentage has changed. Using the 2023 records, she collects a random sample of 80 customers who bought cans of soup.

a) Write down the hypotheses that should be used to test Becky's belief.
b) Using a 5% significance level, find the critical region for a two-tailed test.
c) Find the actual significance level of a test based on your critical region.

Of the 80 customers who were selected in the sample, 45 bought cans of soup in packs of six.

d) Complete the test.

1.57 — Question 19. Aaron Farms grows grain for stockfeed. Richard, a farmer, receives bags of grain from Aaron Farms and knows from experience that 5% will be spoilt. Richard decides to test whether another supplier, Baker Farms, will deliver fewer spoilt bags of grain. Richard makes several orders with Baker Farms and takes a random sample of 550 bags of grain.

a) Write down suitable hypotheses to test whether the number of bags of spoilt grain has been reduced.
b) Using a 10% significance level, find the critical region such a test.
c) Of the 550 bags of grain, 13 are found to be spoilt. Complete the test.

1.57 — Question 20. John collects dinosaur models that are sold in sealed egg containers that the collector cannot see through. On average, 8% of the egg containers contain a limited edition gold dinosaur. John purchases 25 of the eggs. This may be considered to be a random sample.

a) Find the probability that

i) exactly one of the eggs contains a gold dinosaur,

ii) at most two eggs contain gold dinosaurs.

b) John buys 30 more eggs from a website. This may also be considered to be a random sample. John believes there is a greater chance of obtaining gold dinosaurs online. John decides to conduct a hypothesis test at the 5% level.

i) Write down suitable hypotheses for the test.

ii) Find the critical region for the test.

iii) John opens the eggs to find there are four golden dinosaurs. Complete the test.